Organisms and their environment

30 questions: 15 multiple choice, 15 structured.

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1 · Part 1 Core Energy flow [1]

What is the principal source of energy entering most biological systems?

A mineral ions in the soil

B the Sun

C heat released by decomposers

D carbon dioxide in the air

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2 · Part 1 Core Food chains and food webs [1]

Which organism is a producer in a grassland?

A a rabbit

B a hawk

C a fungus growing on dead grass

D grass

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3 · Part 1 Core Food chains and food webs [1]

In the chain grass → grasshopper → lizard → hawk, what is the lizard?

A secondary consumer

B producer

C primary consumer

D quaternary consumer

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4 · Part 1 Core Food chains and food webs [1]

Which feeding description identifies a decomposer?

A making glucose directly from sunlight

B eating only living green leaves

C obtaining energy from dead or waste organic material

D absorbing mineral ions from the soil

5 · Part 1 Core Food chains and food webs [1]

One species of mammal eats only other animals. How is this mammal described?

A herbivore

B producer

C photosynthetic organism

D carnivore

6 · Part 1 Core Food chains and food webs [1]

Why is a food web more informative than one food chain for many habitats?

A It shows that each animal has only one food source.

B It shows interconnected feeding relationships.

C It excludes producers.

D It shows that energy cycles endlessly.

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7 · Part 1 Core Food chains and food webs [1]

Humans remove most predatory fish from a lake. Their main prey are small fish that eat water fleas. What immediate change is most likely, all else unchanged?

A. a decrease in small fish caused by more predation

B. an increase in water fleas

C. an increase in small fish

D. no change in the number of small fish

8 · Part 1 Core Food chains and food webs [1]

A single large tree supports thousands of small herbivores. Why does a pyramid of biomass represent this food chain better than a pyramid of numbers?

A. It accounts for the large differences in organism size.

B. It assumes every organism has the same mass.

C. It always measures energy transferred per year.

D. It ignores producer material.

9 · Part 1 Extended Food chains and food webs [1]

Which process in a living consumer transfers energy from the consumer to the environment?

A. digestion

B. respiration

C. growth

D. photosynthesis

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10 · Part 1 Extended Food chains and food webs [1]

Why do food chains usually contain fewer than five trophic levels?

A Energy increases at every level.

B Producers cannot be eaten.

C Less energy is available at each successive level.

D All organisms are the same size.

11 · Part 1 Extended Food chains and food webs [1]

Why can feeding crop plants directly to humans be more energy efficient than feeding them first to cattle?

A Cattle store all of the energy in the crops they eat.

B Humans digest plant material more efficiently than cattle do.

C Crop plants do not respire, so they lose no energy.

D It avoids energy losses at the cattle trophic level.

12 · Part 1 Extended Nutrient cycles [1]

Which process converts nitrogen gas into nitrogen compounds that can enter living systems?

A nitrogen fixation

B denitrification

C nitrification

D decomposition

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13 · Part 1 Core Populations [1]

Which description is a population?

A every species in a forest

B all abiotic factors in a pond

C all rabbits living in one meadow at the same time

D all herbivores on Earth at any time

14 · Part 1 Core Populations [1]

Which group best represents a community?

A all populations of different species in a pond

B all frogs of one species in a pond

C the temperature and oxygen concentration only

D all water molecules in a lake

15 · Part 1 Core Populations [1]

A new disease spreads in a crowded deer population. What is its likely direct effect on population growth?

A an increase, because the survivors reproduce faster

B a reduction, because more deer die or reproduce less successfully

C no effect, because disease only affects old deer

D an increase, because the disease kills the deer's predators

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16 · Part 1 Core Energy flow [4]

Sunlight enters a grassland food chain.

(a) Describe how the energy first enters living organisms. [2]

(b) Describe how this energy moves through the animals in the food chain and where it is eventually transferred. [2]

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17 · Part 1 Core Food chains and food webs [4]

Pondweed is eaten by snails. Snails are eaten by fish, which are eaten by herons.

(a) Write out the food chain, using arrows to show the direction of energy transfer. [2]

(b) Identify the producer and tertiary consumer. [2]

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18 · Part 1 Core Food chains and food webs [4]

In a meadow, voles eat grass and seeds; owls eat voles; foxes eat voles and rabbits; rabbits eat grass.

(a) Explain one likely effect on owls if vole numbers fall. [2]

(b) Explain why foxes may be less affected than owls in this food web. [2]

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19 · Part 1 Core Food chains and food webs [4]

A hare eats leaves and a wildcat eats hares.

(a) Explain why the hare is both a consumer and a herbivore. [2]

(b) Explain why the wildcat is a carnivore and state its trophic level in this chain. [2]

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20 · Part 1 Core Food chains and food webs [4]

Fallen leaves are broken down by fungi, while nearby green plants grow.

(a) Describe how the fungi obtain energy and classify their feeding role. [2]

(b) Contrast how green plants obtain organic nutrients and energy. [2]

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21 · Part 1 Core Food chains and food webs [4]

One woodland contains 12 mature trees and 2400 leaf-eating insects. Each tree has far more biomass than one insect.

(a) Explain why numbers alone may give a misleading impression of the producer level. [2]

(b) Explain two advantages of using biomass in this example. [2]

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22 · Part 1 Extended Food chains and food webs [4]

An ecologist measures annual energy transfer at each trophic level.

(a) Explain why energy transfer is a useful basis for comparing trophic levels. [2]

(b) State two reasons why not all energy in one trophic level reaches the next. [2]

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23 · Part 1 Extended Food chains and food webs [4]

In one year, the plants in a field contain 18 000 kJ of energy. The herbivores that feed on these plants store 2700 kJ of this energy in new tissue.

(a) Calculate the percentage of the energy in the plants that is transferred to new herbivore tissue. [2]

(b) Explain two reasons why the rest of the energy is not stored in new herbivore tissue. [2]

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24 · Part 1 Extended Food chains and food webs [4]

A farm can produce edible beans or feed the beans to animals and use the animals as food.

(a) Explain why the bean-to-human route can support more people from the same crop energy. [2]

(b) Explain why adding more trophic levels reduces the population supportable at the top. [2]

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25 · Part 1 Core Food chains and food webs [4]

An introduced insect eats the main seeds used by native seed-eating birds. A predator eats those birds.

(a) Explain a possible effect on the native birds. [2]

(b) Explain how the effect could spread to another trophic level. [2]

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26 · Part 1 Core Nutrient cycles [4]

Carbon moves between a forest and the atmosphere.

(a) Describe how carbon enters plants and then herbivores. [2]

(b) Describe two processes that return carbon dioxide to the atmosphere. [2]

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27 · Part 1 Extended Nutrient cycles [4]

Dead organisms supply nitrogen compounds to soil. Some nitrogen is later lost from waterlogged soil.

(a) Describe how decomposers and nitrifying bacteria make nitrogen available to plants. [2]

(b) Describe nitrogen fixation and denitrification, distinguishing their directions. [2]

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28 · Part 1 Core Populations [4]

A wetland contains reeds, fish, insects and microorganisms interacting with water and soil.

(a) Distinguish the reed population from the wetland community. [2]

(b) Explain why the wetland is an ecosystem. [2]

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29 · Part 1 Core Populations [4]

A rabbit population grows while food is plentiful. Later, food becomes scarce and fox numbers rise.

(a) Explain how food scarcity can limit population growth. [2]

(b) Explain how increased predation can also limit growth. [2]

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30 · Part 1 Extended Populations [4]

A bacterial culture initially grows slowly, then rapidly, and finally remains approximately constant in size.

(a) Explain the slow initial phase and the later rapid phase. [2]

(b) Explain why the population can stop increasing. [2]

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